Predicting Airbnb Booking Success: San Francisco Market Analysis

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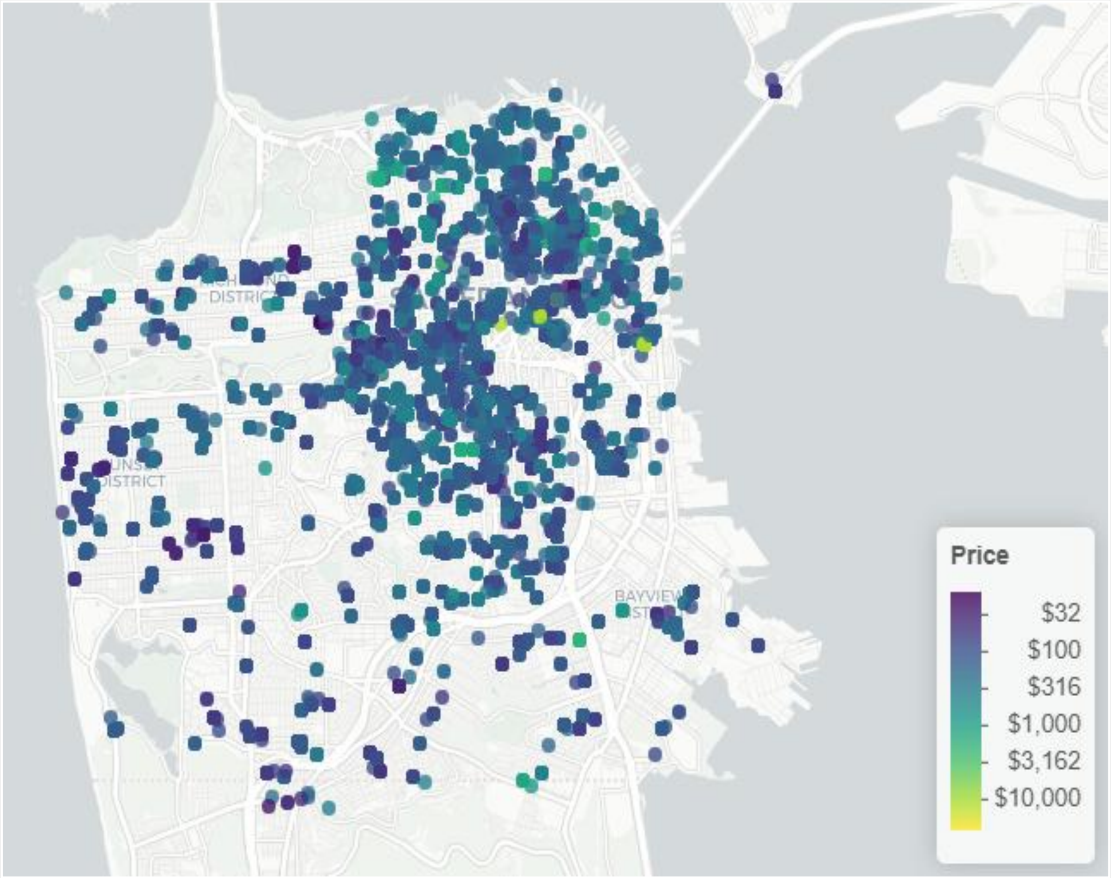
Executive Summary

Map of San Francisco Airbnb Bookings



*Analysis used machine learning models to predict booking success

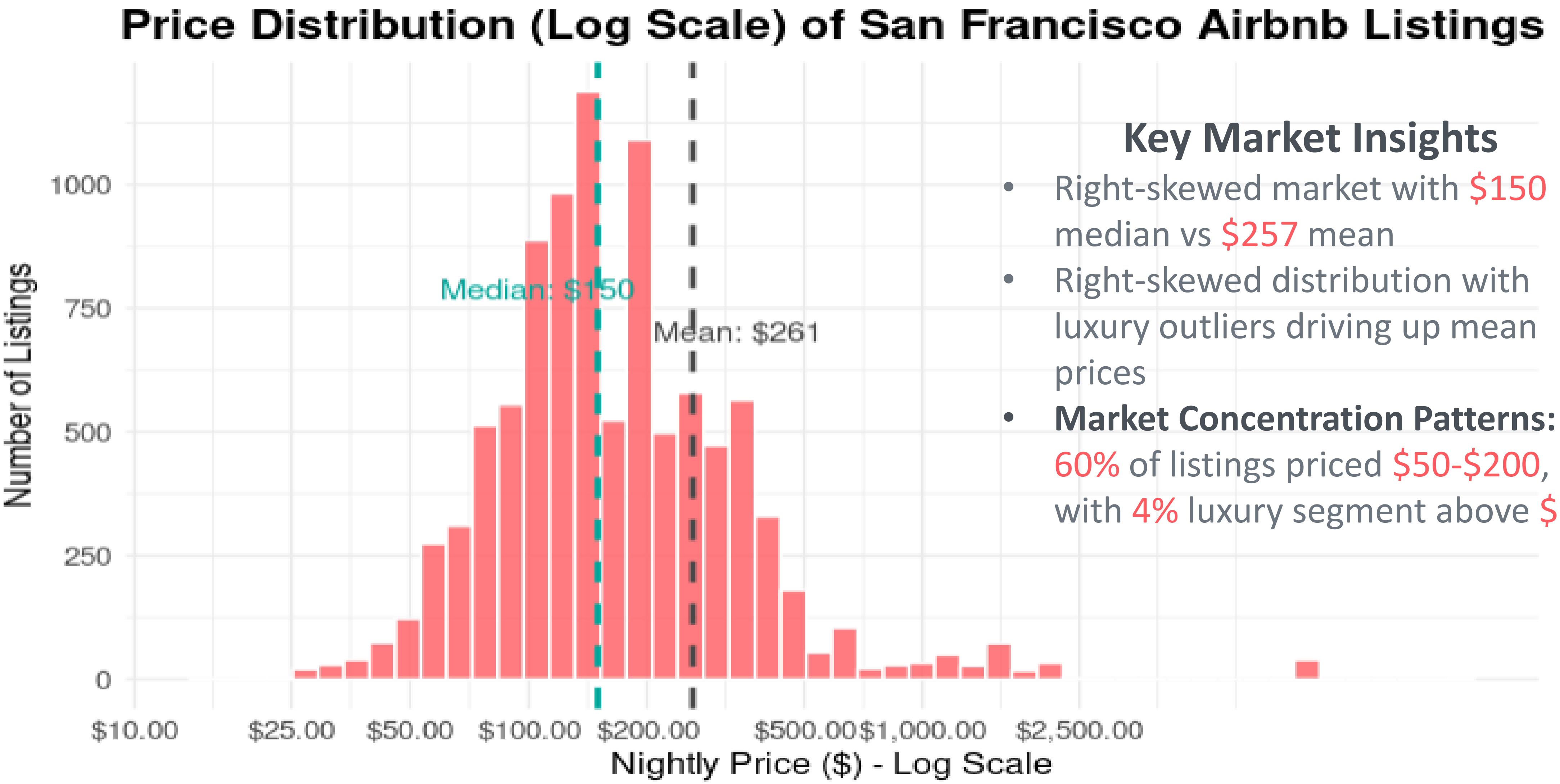
Dataset Snapshot:

- 12,947 Total Properties
- 3,768 High-Booking Properties (**29%**)
- **$257.21** Mean Price
- **$150** Median Price

Research Objectives:

- Identify key factors predicting high booking rates
- Understand price distribution and market dynamics
- Develop predictive models for property performance
- Provide actionable investment insights

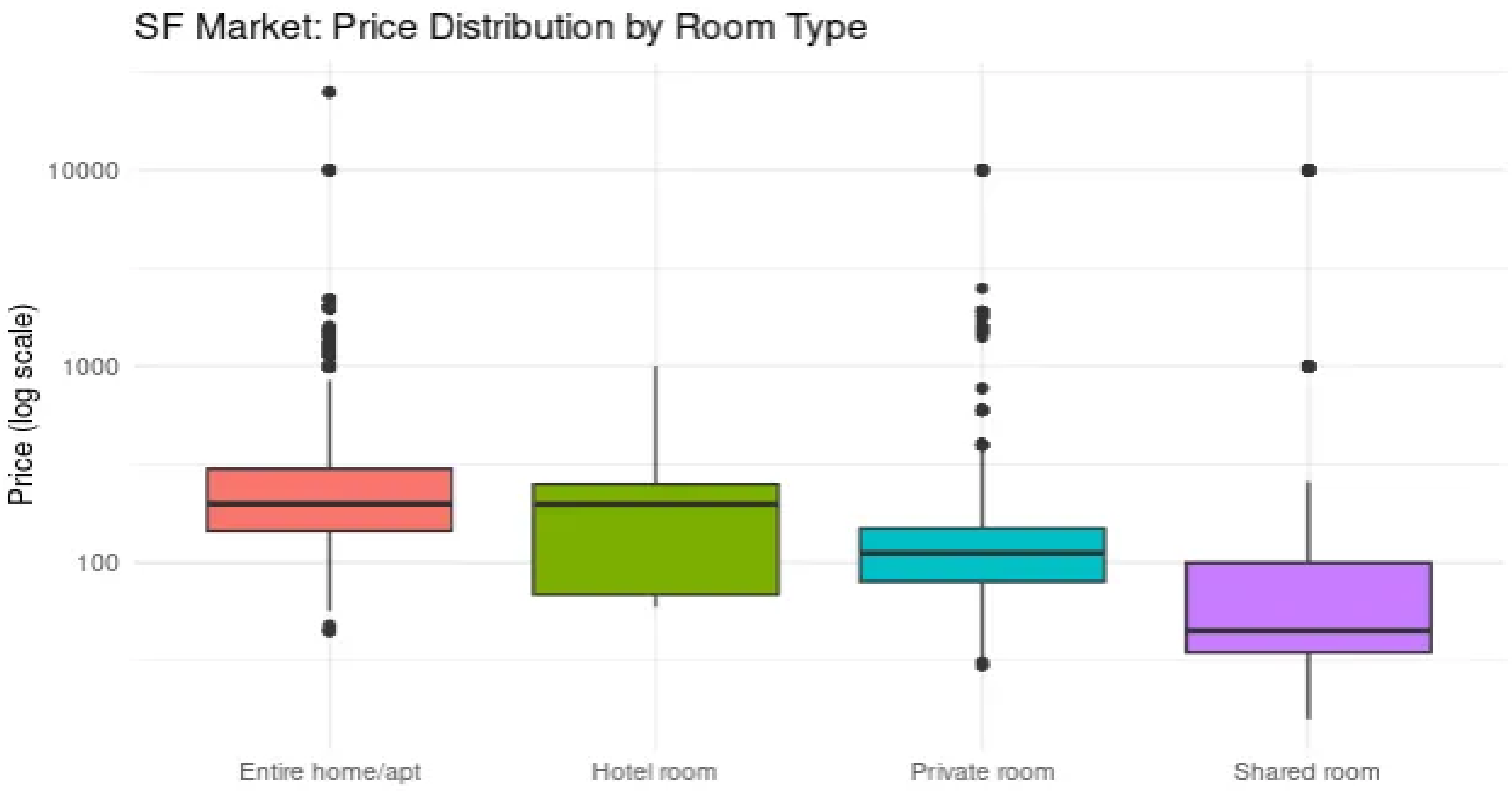
Price Distribution Analysis



Key Market Insights

- Right-skewed market with $150 median vs $257 mean
- Right-skewed distribution with luxury outliers driving up mean prices
- **Market Concentration Patterns:** 60% of listings priced $50-$200, with 4% luxury segment above $500

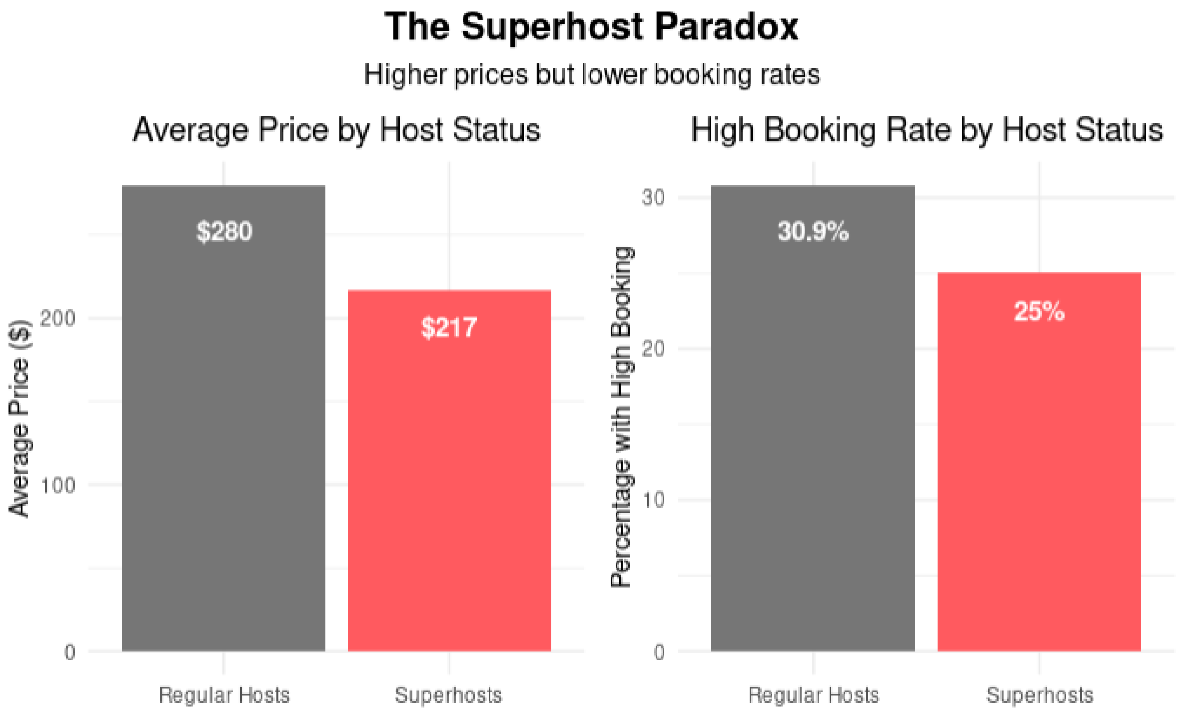
Room Type Analysis & Price Hierarchy

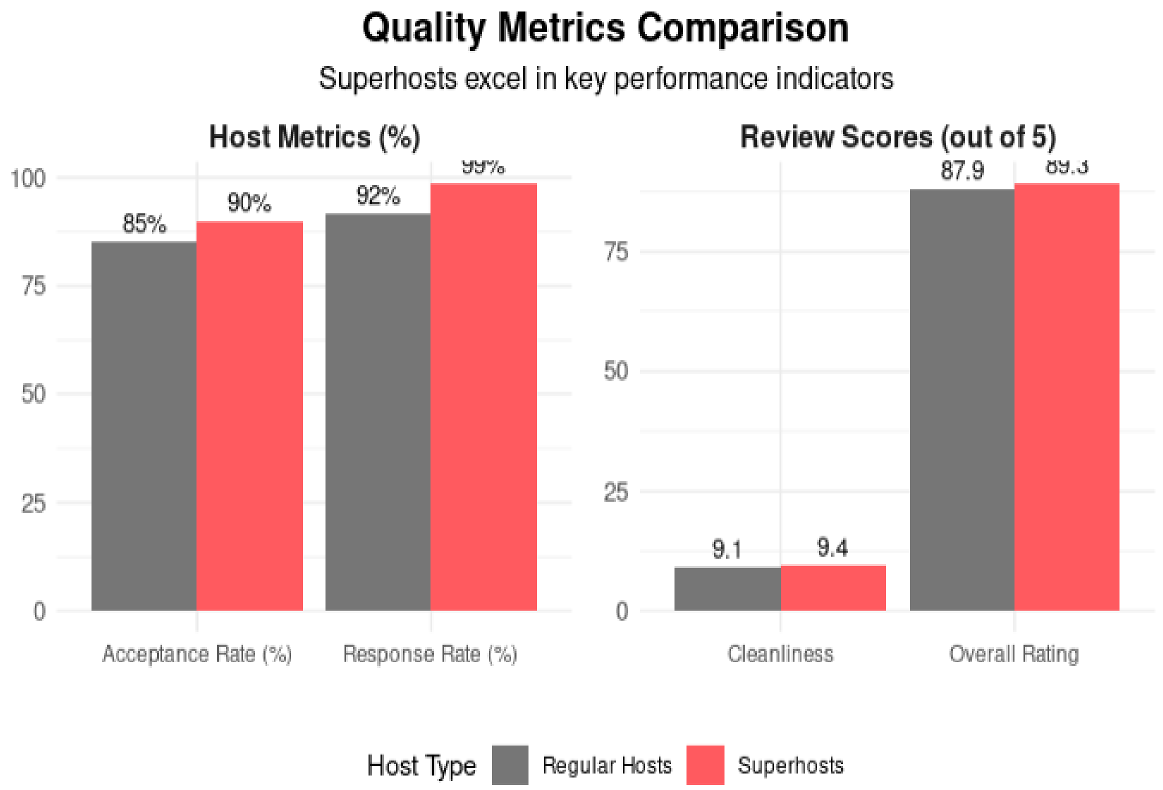


Entire Home:	Hotel Room:	Private Room:	Shared Room:
$200	~$175	~$125	$50

- **Privacy drives pricing power:** Clear 4x price difference from shared rooms ($50) to entire homes ($200)
- **Entire homes** offer highest revenue potential with widest pricing flexibility, while **shared rooms** provide stable, standardized income
- **Luxury market** exists across all segments with outliers reaching $10,000+, indicating premium positioning opportunities regardless of property type

Superhost Strategy: Value Over Volume





Key Findings

- Lower pricing strategy: $217 vs $280 for regular hosts
- Quality over quantity: 25% high-booking rate vs 31% (selective approach)
- Superior service metrics: 99% response rate vs 92%

Variable Selection Methodology

airbnb

Raw Dataset
12,947 properties

→

Data Preprocessing
Cleaned dataset

→

Model Variables
32 features

→

ML Models
95% AUC

Property Features

- accommodates
- bedrooms
- bathrooms
- beds

Host Attributes

- superhost_status
- response_rate
- acceptance_rate
- profile_pic

Pricing Variables

- daily_price
- weekly_price
- monthly_price
- security_deposit

Review Scores

- cleanliness
- location
- communication
- accuracy

Data Preprocessing Steps:

✓ Removed text variables and unique identifiers

✓ Retained 32 numerical and categorical features

✓ Handled missing values through imputation

✓ Selected features based on business relevance and statistical significance

Removed text variables

Handled missing values

Feature selection

Algorithm Performance Results

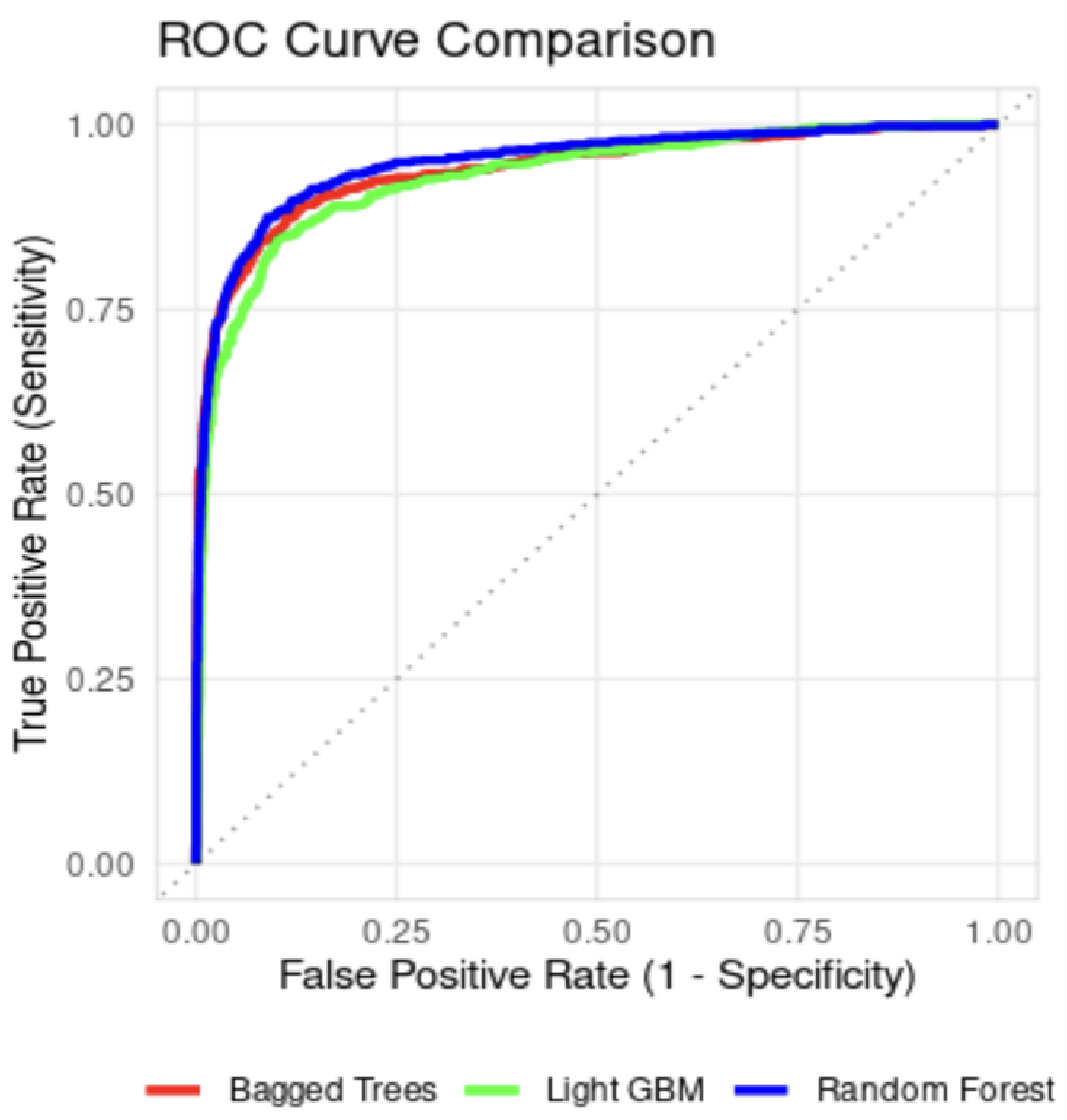
Initial Results (Raw Data):

- Logistic Regression: 59% AUC
- Random Forest: 73% AUC
- Bagged Trees: ~70% AUC (estimated)
- Light GBM: 63% AUC

After Data Preprocessing:

- Random Forest: 95% AUC **(+22%)**
- Bagged Trees: 94% AUC **(+24%)**
- Light GBM: 93% AUC **(+30%)**

Key Insight: Random Forest achieved best performance at 95% AUC



Key Predictors from Logistic Regression

	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	1.452833e+02	1.474277e+02	0.9854551	3.244006e-01
host_is_superhost	1.330138e-01	6.320540e-02	2.1044689	3.533756e-02
accommodates	-7.848034e-02	2.419056e-02	-3.2442544	1.177585e-03
bathrooms	-3.538846e-02	4.534557e-02	-0.7804172	4.351453e-01
bedrooms	5.008422e-01	4.908671e-02	10.2032147	1.918163e-24
beds	-4.894593e-02	3.018906e-02	-1.6213133	1.049505e-01
extra_people	-6.129171e-03	1.543832e-03	-3.9701017	7.184196e-05
guests_included	3.305525e-03	2.650666e-02	0.1247055	9.007567e-01
host_acceptance_rate	8.489970e-03	6.920728e-02	0.1226745	9.023648e-01
host_has_profile_pic	7.439168e-01	2.355681e-01	3.1579687	1.588726e-03

Model Interpretation Results:

- LASSO regularization retained 34 most predictive features
- 10+ variables statistically significant ($p < 0.05$)
- Interpretable coefficients enable business decision-making

Key drivers: bedrooms, host profile, pricing strategy

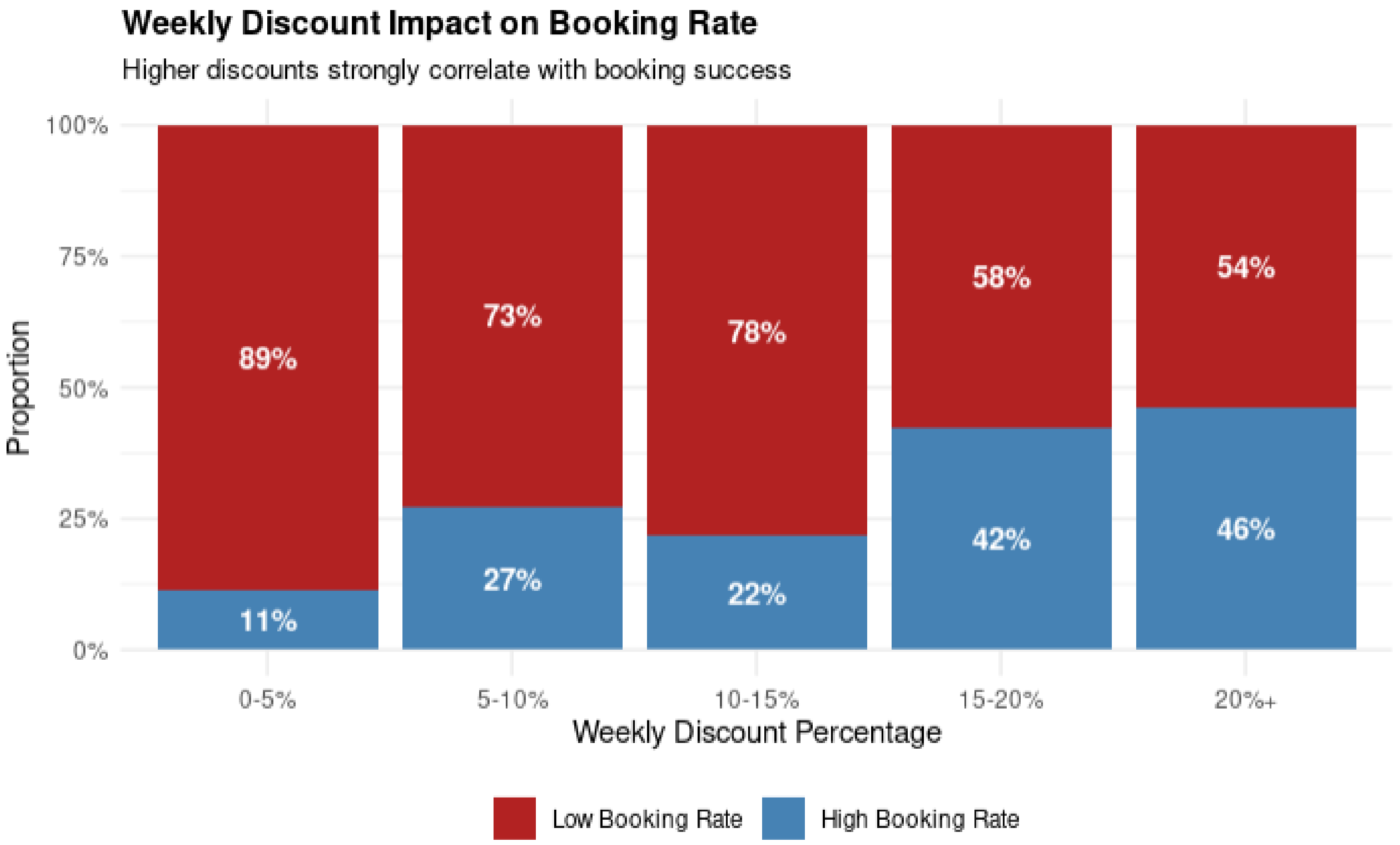
Example Coefficient Interpretation:

exp(0.50) = 1.65 for bedrooms coefficient

Translation: Each additional bedroom increases the odds of high booking rates by 65%, holding all other factors constant.

Weekly Discount Strategy

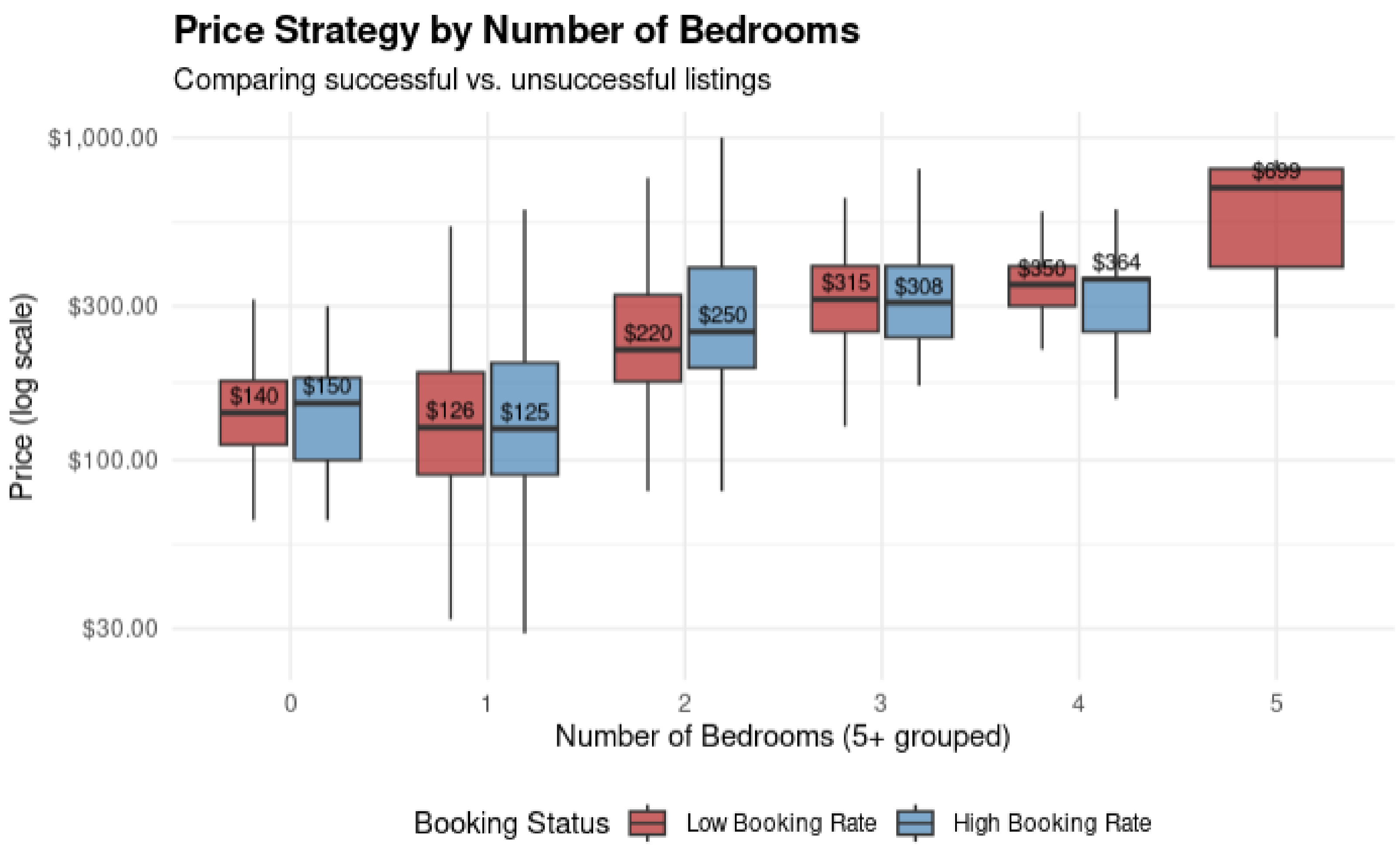
High-booking properties use **aggressive** weekly discounts: 14.3% vs 5% industry standard



- **Discount-driven success:** Properties offering 12-15% weekly discounts significantly outperform those with minimal discounting
- **Competitive pricing strategy:** Price 5-10% below comparable listings while using generous discounts to drive occupancy
- **Value perception wins:** Data shows discount strategy beats rock-bottom pricing for sustained booking success

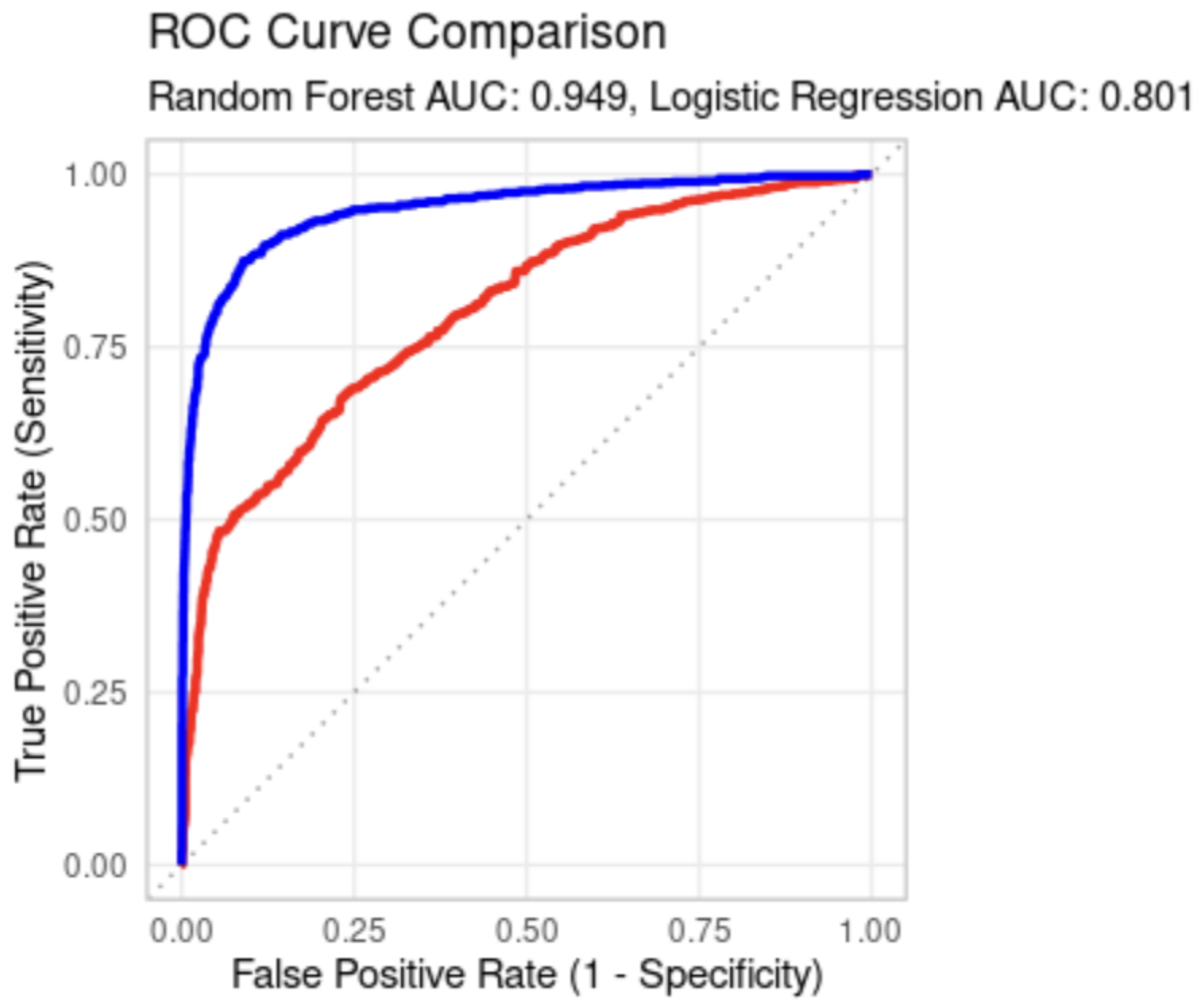
Location & Bedroom Pricing Patterns

Pricing sweet spots vary dramatically by **property size** and **location**



- **Location premiums work:** Central neighborhoods command 30-40% price premiums while maintaining high booking rates
- **Small property competition:** Properties with 0-2 bedrooms show smallest pricing gaps between high/low performers - most competitive segment
- **Size-based strategy:** Each bedroom count has optimal pricing thresholds - larger properties have more pricing flexibility

Model Performance Achievement



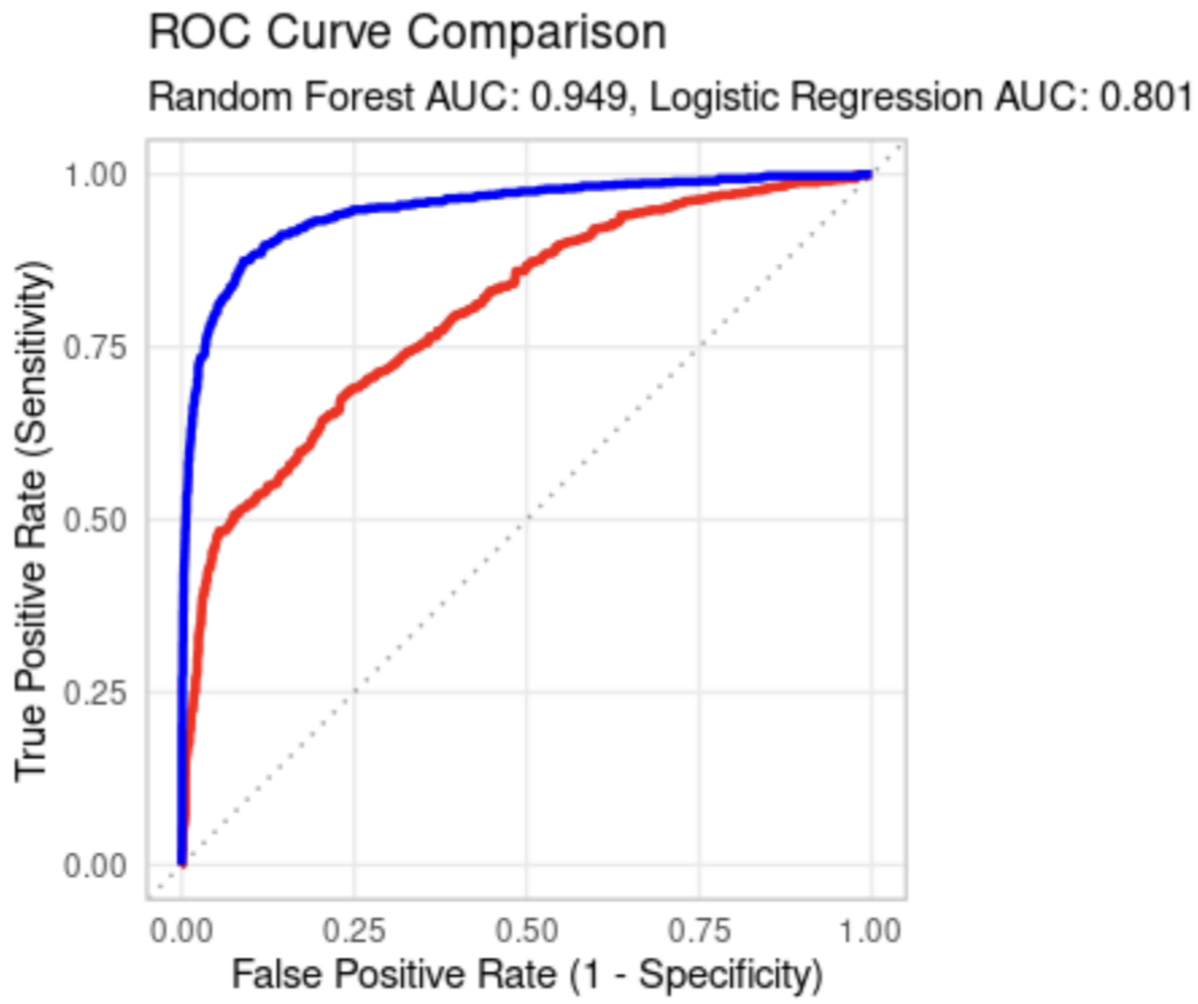
Random Forest achieved **95% AUC**

Before:
Raw data
(73%)

After:
Cleaned data
(95%)

*22-point Improvement Unlocks Reliable Predictions

Future Model Enhancements



What To Add To Future Model:

- Transit proximity scores (BART/Muni distance)
- Seasonal demand patterns
- Competitive pricing indices
- Amenity clustering analysis

Investment Strategy Framework

Property Type Hierarchy

Entire Homes
$200

Private Rooms
$125

Shared Rooms
$50

Target Property Criteria:

- Focus on entire homes or private rooms
- Implement substantial weekly discounts
- Prioritize properties with instant booking

Pricing Strategy Formula

Base Price
5-10% Below Market

+

Weekly Discounts
12-15% Off

=

Result
Optimal Booking Success

San Francisco Market Opportunity

Executive Briefing on Market Entry & Positioning

12,947
Total Properties
Active Listings

29%
High-Booking Rate
vs Industry Average

$150/$257
Median/Mean Price
Right-Skewed Market

29.5%
Superhost Success
vs 19.4% Regular

Market Insight:
$707 Standard Deviation =
Diverse Pricing Opportunities

Three Strategic Pillars for Market Success

Data-Driven Approach to Maximize ROI

Value-Based Pricing

Price 5-10% below market while maintaining quality standards

93% Optimal Rate

Discount-Driven Occupancy

Aggressive weekly discounts outperform rock-bottom pricing

14.3% Weekly

Location Premiums

Central neighborhoods command 30-40% price premiums

+40% Central SF

SAN FRANCISCO

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Thank You!

Questions or Comments?